

HW04 Main Report — Automation Testing

Student information

Field	Value
Student	23127379 — Thái Minh Huy
Assignment	HW04 — Automation Testing
Repository	https://github.com/AkiraTomori/eshop-sut
Framework	Playwright 1.62.0 with TypeScript
Browsers	Chromium, Firefox, WebKit
Scope	Browser UI automation only

1. Feature selection and automation scope

The same three web features selected in HW02 were automated. API/status/database clauses were excluded and documented in each automation review rather than simulated through Playwright request APIs.

Pool	Feature	Automated UI TCs	Latest evidence run	Browser executions
A	FR-06 — Product Detail View	22	Run #1	66
B	FR-08 — Checkout	14	Run #6	42
C	FR-15 — Product Management	25	Run #2	75
Total	3 features	61		183

Each feature exceeds the minimum of 12 UI test cases. The authoritative TC selections, hybrid-clause limitations, API exclusions, and source discrepancies are recorded in the pool-specific automation reviews.

2. AI-first workflow and human review

The work followed the repository's gated workflow: shared Playwright setup, one-feature script generation, script review and correction, isolated three-browser execution, failure classification, and HITL audit sign-off. The AI loaded the current FR's HW2 test cases, SRS, and known-bug report before generating each suite. External data, fixture lifecycle, BasePage-derived page objects, stable locators, and spec-correct expectations were reviewed at every FR gate.

Material corrections included:

- FR-06: replaced one-time boolean assertion branches with web-first assertions, removed a helper that masked the defective first Add-to-Cart click, and separated UI-observable clauses from API-only clauses.

- FR-08: corrected Profile controls to verified placeholder locators, reordered Profile/cart setup after discovering that cart state is not reload-persistent, and replaced an invalid accessible-name assumption for checkout list items.
- FR-15: corrected generated TC mappings and scope partitions, documented a destructive update case as HITL-only, and classified unrelated price-format and disputed tab-order assertions as test/source issues instead of product bugs.

The signed audit history is available in `Infrastructure-AI-Audit.md` and each pool's `FR##-AI-Audit.md`. Earlier partially accepted FR-08 runs remain in the history; later reviewed corrections and accepted Run #6 supersede those unsuccessful cycles.

3. Automation design

3.1 Data-driven implementation

All URLs, credentials, input values, labels, boundaries, and expected results are supplied outside the test bodies through `test-environment.json` and pool-specific JSON files:

- `Pool-A_FR06/fr06-test-data.json`
- `Pool-B_FR08/fr08-test-data.json`
- `Pool-C_FR15/fr15-test-data.json`

Specs import `test` and `expect` only from `fixtures/eshop.fixture.ts`. The fixture creates and closes a fresh authenticated context per test. Feature page objects extend `pages/base.page.ts`; they contain user-intent operations and verified locators, not mutable test data or assertions.

3.2 Assertions and isolation

Across the suites, assertions include `toHaveURL`, `toBeVisible`, `not.toBeVisible`, `toHaveText`, `toContainText`, `toHaveValue`, `toHaveCount`, `toHaveAttribute`, `toHaveCSS`, `toBeFocused`, and `expect.poll`. Tests use UI-based setup and cleanup and contain no request fixture, direct API/database action, response interception, `page.evaluate()` assertion, or wait longer than 500 ms.

3.3 Multi-browser reporting

`playwright.config.ts` defines Chromium, Firefox, and WebKit projects and starts or reuses the Backend, Web, and Web Admin services. Every retained browser report displays `Run by: 23127379` and records an ISO run timestamp in Playwright metadata. Each pool contains a full overview and isolated browser reports under `playwright-report/`.

4. Latest execution results

Feature	Chromium	Firefox	WebKit	Aggregate
FR-06	7 passed / 15 failed	7 / 15	7 / 15	21 passed / 45 failed
FR-08	1 passed / 13 failed	1 / 13	1 / 13	3 passed / 39 failed
FR-15	1 passed / 24 failed	1 / 24	1 / 24	3 passed / 72 failed
Total	9 / 52	9 / 52	9 / 52	27 passed / 156 failed / 183 total

Nonzero exits are retained evidence, not incomplete runs. FR-06 and FR-08 failures were classified predominantly as genuine spec-correct product failures. FR-15 contains 57 genuine product-failure results

and 15 results caused exclusively by two test/source issues. The cumulative run summaries also preserve infrastructure-only attempts with zero discovery; these are not substituted for the latest evidence runs.

5. Defect results

The consolidated report records 28 distinct genuine defects: 19 known HW2 defects reproduced and nine automation-discovered defects. All 156 failed browser results are classified in the detailed pool reports; failures are not automatically treated as bugs.

Feature	Genuine defects	Known	New	Test/source issues
FR-06	8	7	1	1 secondary observation
FR-08	8	6	2	0 in latest run
FR-15	12	6	6	2
Total	28	19	9	3

All 28 known and automation-discovered defects link to verified GitHub Issues. The nine automation-discovered defects are Issues 59–67, and their detailed reports include the submitted evidence. See `bug_report.md` and the three detailed `fr##-bug-report.md` files.

6. Cases not fully automated

- Direct HTTP, authorization-header, malformed-token, response-status, tampered-payload, and database verification cases are listed as `Out of HW4 scope – API testing` in the respective reviews.
- Hybrid cases automate only their explicit UI path and UI-observable outcome; API/database clauses are listed as uncovered.
- FR-15 EP-003 is not counted because its update behavior mutates unrelated shared products and the current UI cannot safely restore their identities. It requires a disposable database snapshot under HITL control.

7. Demo video

The repository declares the unlisted YouTube playlist <https://www.youtube.com/playlist?list=PLFKmUDyIY8sc>, with six linked workflow parts listed in `README.md`. Before submission, the student must confirm that the submitted video evidence totals at least five minutes, is narrated in Vietnamese, demonstrates an end-to-end multi-browser run and HTML report, explains at least one correction, and shows face-cam or terminal `whoami` and `hostname`. Those content requirements cannot be established from repository metadata alone.

8. Reusable Agent Skills

The repository contains reusable skills for Playwright infrastructure, data-driven script generation, script review, three-browser evidence, failure classification, and AI-audit logging under `.agents/skills/`. They enforce the browser-UI-only boundary, sequential FR gates, external test data, BasePage-derived POM structure, isolated reports, and HITL sign-off.

9. Git history

`git-commit-log.txt` records nine commits that modify `HW4.spec.ts` files. This satisfies the eight-commit count but spans only 2026-07-31 and 2026-08-01—two calendar days, not the required four. This is disclosed as a submission-compliance gap and must not be corrected by rewriting commit history.

10. Provisional self-assessment

Criterion	Maximum	Provisional score	Rationale
Feature A — FR-06	25	23	Complete suite, evidence, classification, and GitHub Issue links
Feature B — FR-08	25	23	Complete corrected evidence, classification, and GitHub Issue links
Feature C — FR-15	25	21	Complete evidence and Issue links, with two classified test/source issues
Demo video	15	9	Playlist supplied; duration/content/authorship require human confirmation
Agent Skills	10	9	Reusable gated workflow and audit tooling are present
Total	100	85	Provisional filename grade: 085

The student should confirm or change this self-assessment before naming the submission archive.

11. Submission readiness

Completed locally:

- Three data-driven Playwright suites and external JSON files
- Three-browser reports for all three features, including full-FR overviews
- Automation reviews, cumulative run summaries, detailed bug reports, consolidated bug index, and signed AI audits
- Main report, AI critique, README summary, spec-focused Git commit log, and consolidated Markdown/PDF AI Audit appendix

HITL actions still required before Moodle submission:

- Confirm the playlist/video satisfies duration, Vietnamese narration, end-to-end report demonstration, correction narration, and authorship evidence
- Confirm the provisional self-assessed grade and final archive filename
- Inspect the generated PDFs and build the ZIP after the remaining HITL checks are resolved
- Acknowledge that the existing qualifying commit history spans two days and cannot satisfy the four-day rule retroactively without falsification

References

- HW04 assignment: `2026.HW04.Automation Testing_En.md`
- HW2 canonical sources: `../HW2/Pool-A_FR06_ProductDetailView`, `../HW2/Pool-B_FR08_Checkout`, and `../HW2/Pool-C_FR15_ProductManagement`
- Shared SRS: `../HW2/agents/context/eshop-srs.md`

- Playwright documentation: <https://playwright.dev>